

SPRING CORE AND MAVEN

EXERCISE 2

Objective

To implement Dependency Injection using Spring IoC and XML configuration

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {

        this.bookRepository = bookRepository;

    }

    public void displayService() {

        System.out.println("Book Service is working.");

        System.out.println("Calling Book Repository...");

        bookRepository.displayRepository();

    }

}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

        xsi:schemaLocation="
```

<http://www.springframework.org/schema/beans>

<https://www.springframework.org/schema/beans/spring-beans.xsd>>

```
<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>
```

```
<bean id="bookService"
      class="com.library.service.BookService">
  <property name="bookRepository"
            ref="bookRepository"/>
</bean>
```

```
</beans>
```

MainApp.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
```

```
public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.displayService();
    }
}
```

```
}  
  
}
```

Output

The application executed successfully and displayed the following output:

```
Jun 29, 2026 03:18:45 PM org.springframework.context.support.AbstractApplicationContext prepareRefresh  
INFO: Refreshing org.springframework.context.support.ClassPathXmlApplicationContext@2b193f2d  
  
Jun 29, 2026 03:18:45 PM org.springframework.beans.factory.xml.XmlBeanDefinitionReader loadBeanDefinitions  
INFO: Loading XML bean definitions from class path resource [applicationContext.xml]  
  
Book Service is working.  
Calling Book Repository...  
Book Repository is working.  
  
Jun 29, 2026 03:18:45 PM org.springframework.context.support.AbstractApplicationContext doClose  
INFO: Closing org.springframework.context.support.ClassPathXmlApplicationContext@2b193f2d  
  
<terminated> LibraryManagementApplication [Java Application]
```